PULSE — Regime-Conditional Crude Spread Engine

Methodology & Walk-Forward Results · Generated 2026-06-11 12:34 UTC · Reproduce: `python -m backend.research.walkforward` · `python -m backend.research.methodology_pdf`

1. Problem

Identify when crude calendar spreads (front carry, mid-curve carry, front butterfly) deviate from their *regime-conditional* fair value. Output a single ranked opportunity per day with full receipts: fair, 80% band, residual z-score, top drivers, historical analogs.

2. Universe

6 instruments — 3 Brent (M1-M2, M3-M6, front fly M1-2×M2+M3) + 3 WTI mirror instruments. Brent legs from /Data C1-C31 daily settlements (2016-2026, 2,712 trading days). WTI legs from synthesised daily settlements (last 1-min mid per session, 2021-2026, 1,676 trading days — flagged ESTIMATE pending the real exchange file).

3. Regime grids — composite + pooled

Composite (Sprint 3): 27 cells = **CURVE** {CONTANGO / NEUTRAL / BACK on Brent M1-M12 at -$2, +$5} × **INVENTORY** {LOW / AVG / HIGH on US crude stocks vs 5y seasonal at ±4%} × **VOL** {CALM / NORMAL / STRESSED on Brent 20d realised vol at 20%, 35%}. Hard thresholds — mentor can answer 'why this regime?' in one line. ~10-15 cells/spread have usable n; the rest are economically implausible.

Pooled (Phase 2.5): 3 cells/spread = curve axis only {CONTANGO / NEUTRAL / BACK}. Same thresholds; inventory + vol still surface on the UI as regime context but no longer fragment the training set. Average ~770 rows per cell (vs ~150 composite) — explicitly addresses the Sprint 4 finding that per-cell data was too thin to beat a 252-d rolling z.

4. Feature set (Phase 2.8.2 expansion)

20 Brent features per cell (was 11): *Curve shape* — M1-M12, its square, and **curvature** (M1 – 2×M6 + M12) for fly-specific information. *Front state* — Brent close + 5d return + 20d realised vol. *Inventory* — US crude vs 5y seasonal AND **inventory surprise** (weekly Δ minus its 4-week MA). *Seasonality* — sin/cos of day-of-year. *Lagged spreads* — M1-M2 / M3-M6 / fly lag-1.

Phase 2.8.2 alpha features (new) — **COT managed-money percentile** (rolling 156-week, contrarian crowding), **3-2-1 crack** + **gasoline crack** (refining-margin pressure on the front), **WTI-Brent spread** (Atlantic-basin arb), **real rate** (DGS10 – 5y TIPS breakeven; storage-cost driver), **OVX/VIX ratio** (crude vs equity vol risk-premium), and **days to front-month expiry**. WTI cells append 5 more (WTI close, M1-M12, 3 lag-1 spreads). The target spread's own lag is removed to avoid self-leakage.

5. Per-cell model competition

We *do not* pick a single regression family a priori. For each (spread, regime) cell, **seven candidates** compete on 5-fold TimeSeriesSplit cross-validated R^2 (Phase 2.8.1 expansion):

Linear: Ridge (shrinks coefficients — good with correlated predictors), Lasso (zeroes out noise features — good when 3-4 signals dominate), ElasticNet (L1+L2 blend — mixed correlation), Huber (robust loss — COVID-2020 / Russia-2022 outliers).

Gradient boosting (Phase 2.8.1): XGBoost, LightGBM, CatBoost — shallow trees with strong regularisation that capture non-linear interactions between features. Win in cells where linear leaves alpha on the table.

Winner = max mean CV R^2, simplicity tiebreak within 0.005: linear families rank above boosters (ElasticNet > Lasso > Ridge > Huber > XGBoost > LightGBM > CatBoost), so interpretability wins when CV scores are close. Quantile regressors (p10/p50/p90) are fit separately for the 80% confidence band.

6. Trading rule

At each daily settle: classify today's composite regime → look up winner model for (spread, regime) → predict fair, p10, p50, p90 → z = (actual – fair) / residual_σ from training. Direction = SELL if z > +0.5σ, BUY if z < –0.5σ, else NEUTRAL. Target = p50 (mean reversion). Stop = entry ± 1.5σ. Horizon = 20 trading days.

7. Walk-forward design

Refits every quarter from 2024-Q1 through 2026-Q2 (10 refits). Each refit re-runs the full 4-model competition on data ≤ cutoff, then we walk forward day-by-day to the next cutoff, generating signals. **Phase 2.5 runs BOTH grids in parallel** (composite + pooled) over the same refit dates and the same trading rule, then compares each to a regime-*unaware* 252-day rolling z-score baseline — isolating the contribution of regime conditioning and of grid resolution.

8. Phase 2.6 — gated blend (production rule)

Phase 2.5 surfaced the BACK × {Lasso, Huber} slice as the only (spread × regime × winner) subset where the pooled engine demonstrably beats baseline in walk-forward. **Phase 2.6 codifies that finding as a production rule** and tests it as a third leg alongside composite / pooled / baseline. For each (date, spread): (i) if the pooled candidate has *regime_pooled='BACK'*, *winner_model* ∈ {Lasso, Huber}, AND |*z*| ≥ 0.5σ → take the pooled signal; (ii) else fall back to the 252-day rolling-z baseline. This routing is the exact rule shipped under PULSE_GATED_BLEND=1, so the table on page 2 measures the strategy a trader would actually run.

9. Phase 2.7 — position sizing for the regime leg

Phase 2.6 left max drawdown on the gated blend at −271 (vs −169 baseline) — the cost of concentrating 97 high-Sharpe regime fires on a small number of days. **Phase 2.7 scales the regime-leg notional** while leaving the baseline leg at 1.0, asking whether sizing compresses the DD without giving up the +1.332 regime-leg Sharpe. Three modes simulated end-to-end as a fourth walk-forward leg behind PULSE_GATED_SIZE=<mode>:

full — scale 1.0 (sanity check; identical to gated_blend). **half** — scale 0.5 (uniform risk reduction). **kelly** — per-spread Kelly fraction $f^* = p - (1-p)/b$ computed at each refit boundary from prior closed regime-leg PnLs (p = win rate, b = mean_win / |mean_loss|). Clamped to [0.10, 1.00]; defaults to 0.50 when fewer than 5 prior trades exist. The schedule is expanding-window so the per-spread sample grows over time. Live inference reads the per-spread Kelly at the latest refit boundary from sized_blend_summary.kelly_per_spread_latest.

10. Phase 2.8 — real model + real features

Phase 2.8.1 adds **XGBoost / LightGBM / CatBoost** to the per-cell competition so cells with non-linear interactions aren't forced to fit a linear model. Boosters fit only on cells with ≥80 train rows (smaller samples can't reliably distinguish trees from linears via CV); shallow trees (depth 3-4) and low learning rate (0.05) keep them from memorising. Boosters rank below linears in the tiebreak so interpretability still wins on ties.

Phase 2.8.2 nearly **doubles the feature set** (11 → 20 Brent features) with the alpha-bearing predictors documented in §4: COT crowding, inventory surprise, curvature, refining cracks, real rate, OVX/VIX ratio, WTI-Brent arb, days-to-expiry. The aim is to give every model family — linear and tree — a richer set of predictors so the gated-blend Sharpe lifts by tightening the pooled BACK fair-value estimates the gate rule relies on.

Phase 2.8.3 widened the gate to admit boosters (GATED_WINNERS = {Lasso, Huber, XGBoost, LightGBM, CatBoost}). The widened gate is methodologically correct — the regime leg's reported Sharpe lifted from +0.369 to **+0.888** across 244 fires (was 71) — but the headline blended Sharpe stayed flat (+0.389 -> +0.384) because most of the newly-routed booster trades had baseline-z direction matching pooled-z direction. The widening REASSIGNS attribution rather than CREATING new alpha at the blended level.

11. Phase 2.8.6 — transaction costs

Every Sharpe/PnL number above is **gross** — per-trade transaction cost is not modelled by the walk-forward's spread-move PnL. Phase 2.8.6 layers a defensible cost model on top of the existing trade tape so the table on page 2 reports **NET (after-cost)** Sharpe alongside gross.

Cost model. Per-leg, per-side: $0.0025/bbl commission + clearing + brokerage (~$2.45 per 1,000-bbl contract) + half bid-ask slippage ($0.0050/bbl for front M1/M2, $0.0075/bbl for deferred M3-M6). Round-trip cost = N legs × 2 sides × (commission + half-spread). That gives: **2-leg M1-M2 = $0.030/bbl RT, 2-leg M3-M6 = $0.040, 3-leg fly = $0.050**. Cost scales with sizing_scale on regime rows (half-sized regime trade = half the contracts = half the fees). NEUTRAL trades incur zero cost (no fill).

Cost is applied at the aggregation step only — per-cell CV is on gross fair-value R^2, so winner selection is unaffected and no retraining is required. Live inference does *not* subtract cost at entry; the cost-aware NET metric exists purely for backtest reporting.

Walk-forward results — 2024-2026

Refits: 10 · Records/mode: 3,639 · Horizon: 20 trading days · Universe: 6 spreads · Composite grid: 27 cells · Pooled grid: 3 curve cells · Gated blend (Phase 2.6): regime engine fires on BACK + {Lasso,Huber} + $|z|\geq 0.5\sigma$, else 252d z baseline (regime/baseline split: 258 / 3,380) · Sized blend (Phase 2.7): regime-leg notional scaled by {full=1.0, half=0.5, kelly=per-spread f*}

Overall — composite / pooled / gated blend / baseline

Metric	Composite (27)	Pooled (3)	Gated blend	Baseline 252d z	Δ gated vs base
Signals fired	2281	1992	2154	2082	—
Hit rate	61.6%	64.0%	71.0%	71.6%	-0.6%
Mean PnL ($/bbl)	+0.194	+0.202	+0.176	+0.180	-0.004
Median PnL	+0.090	+0.110	+0.159	+0.164	—
Total PnL	+442.6	+402.4	+379.0	+373.9	—
Sharpe (ann.)	+0.43	+0.44	+0.38	+0.39	-0.00
Max drawdown	-38.9	-46.9	-255.5	-168.9	—

Phase 2.6 finding: even the gated blend can't pull the headline above baseline — the BACK + {Lasso,Huber} slice was strong in Phase 2.5 but it didn't survive a clean walk-forward of the gating rule itself. Sharpe (ann.): composite +0.43, pooled +0.44, **gated +0.38**, baseline +0.39. Gate routing: 258 regime fires / 3,380 baseline fallbacks (7.1% of records hit the engine). The gate doesn't beat baseline in walk-forward. Next: try a wider feature set inside the pooled regression (EIA inventory surprise, COT positioning, OVX, term-structure derivatives), or revisit gating criteria — the Phase 2.5 slice may have been window-specific.

Per-spread Sharpe — composite / pooled / gated / baseline (Δ vs baseline)

Spread	Comp Shp	Pool Shp	Gate Sig	Gate Shp	Base Shp	Δ Gate
brent_fly_123	+0.98	+0.88	253	+1.53	+1.76	-0.23
brent_m1_m2	+1.47	+0.85	345	+0.93	+0.94	-0.00
brent_m3_m6	-0.28	-0.17	419	-0.27	-0.27	0.000
wti_fly_123	+0.89	+0.93	360	+0.88	+0.90	-0.01
wti_m1_m2	+0.24	+0.45	373	+0.26	+0.25	+0.01
wti_m3_m6	+0.31	+0.33	404	+0.30	+0.29	+0.01

Gated-blend slices — which leg fired (regime vs baseline) and winner mix

Gated-blend slice	n	Hit	μPnL	Sharpe
source=baseline	1910	71.4%	+0.136	+0.31
source=regime	244	68.4%	+0.490	+0.89
winner=Huber	351	77.5%	+0.380	+2.06
winner=ElasticNet	1004	67.0%	+0.037	+0.06
winner=LightGBM	87	67.8%	+0.107	+1.19
winner=Ridge	200	67.0%	-0.005	-0.03
winner=Lasso	319	78.4%	+0.320	+0.76
winner=XGBoost	71	57.8%	+0.310	+0.86
winner=CatBoost	122	82.8%	+0.624	+1.35

Phase 2.7 — sized regime leg vs gated full / baseline

Metric	Gated full	Sized 0.5×	Sized Kelly	Baseline 252d z	Δ best vs base

Signals fired	2154	2154	2154	2082	—
Hit rate	71.0%	71.0%	71.0%	71.6%	—
Mean PnL ($/bbl)	+0.176	+0.148	+0.140	+0.180	—
Total PnL	+379.0	+319.2	+301.7	+373.9	—
Sharpe (ann.)	+0.38	+0.35	+0.33	+0.39	-0.00
Max drawdown	-255.5	-264.7	-269.5	-168.9	-86.6
Mean regime scale	1.0	+0.500	+0.385	—	—

Phase 2.7 finding: no sizing mode caught baseline at the headline Sharpe (best gated full +0.38 vs baseline +0.39). Max DD on the best variant: -255.5 vs baseline -168.9 (gated full: -255.5). Latest per-spread Kelly fractions: brent_m1_m2=0.10, brent_m3_m6=0.50, brent_fly_123=0.50, wti_m1_m2=0.50, wti_m3_m6=0.52, wti_fly_123=0.31.

Phase 2.8.6 — NET headline (after transaction costs)

Cost model: per-leg per-side $0.0025 commission + half-spread slippage ($0.0050 front / $0.0075 deferred). RT cost per spread: M1-M2 $0.030, M3-M6 $0.040, fly $0.050.

Mode	Gross / Net Sharpe	Gross / Net µPnL	Mean cost / fire	n fired
Baseline 252d z	+0.39 / +0.30	+0.180 / +0.140	+0.0392	2082
Pooled (un-gated)	+0.44 / +0.35	+0.202 / +0.162	+0.0404	1996
Gated blend (2.8.3)	+0.38 / +0.30	+0.176 / +0.136	+0.0395	2154
Sized half	+0.35 / +0.26	+0.148 / +0.111	+0.0370	2154
Sized kelly	+0.33 / +0.25	+0.140 / +0.103	+0.0365	2154

Spread	Gross Shp (gated)	Net Shp (gated)	Net Shp (base)	Δ Net vs base
brent_fly_123	+1.53	+1.32	+1.55	-0.23
brent_m1_m2	+0.93	+0.86	+0.86	-0.00
brent_m3_m6	-0.27	-0.34	-0.34	0.000
wti_fly_123	+0.88	+0.74	+0.76	-0.02
wti_m1_m2	+0.26	+0.21	+0.20	+0.01
wti_m3_m6	+0.30	+0.21	+0.20	+0.01

Phase 2.8.6 finding: transaction costs erode Sharpe roughly uniformly across modes — baseline drag +0.08, pooled +0.09, gated_blend +0.09. Mean cost per fire sits at $0.039-$0.045/bbl depending on spread mix (boosters fire more on the fly which is the most expensive class). **Best net Sharpe: pooled at +0.351.** Gated blend net Sharpe (+0.297) vs baseline net (+0.301) = Δ -0.004 — the gated blend ties baseline at the net headline. The Phase 2.8.3 'flat headline / improved attribution' verdict holds under costs. **Notable side finding**: the un-gated pooled engine (+0.351 net) now beats both gated_blend and baseline under costs — Phase 2.8.1+2 boosters lifted pooled gross enough that the Phase 2.6 gate (added when pooled was losing) is over-restrictive. Concrete recommendation: A/B test PULSE_REGIME_MODE=pooled against PULSE_GATED_BLEND=1 in paper before changing default.

Caveats & next steps

• **WTI is synthesised** — last-1-min midprice per session, not exchange print. Mentor's real WTI C1-C3 daily file slots in without code changes.

• **Sparse cells (composite only)** — 96/162 composite cells (BACK × HIGH, CONTANGO × LOW) stay empty. The pooled grid is dense by construction (3 cells/spread, average ~770 rows/cell vs ~150 composite).

• **Phase 1 not directly comparable** — Phase 1 is a 9-indicator directional Brent signal; the baseline here (regime-unaware 252d z on spreads) is the cleanest test of regime-conditioning value.

• **Costs modelled (Phase 2.8.6)** — defensible per-spread RT cost ($0.030 M1-M2 / $0.040 M3-M6 / $0.050 fly per bbl) layered on top of gross spread-move PnL. Roll slippage and funding still ignored; live deployment should add a per-spread broker quote where available.

• **Mode switch** — set PULSE_REGIME_MODE=pooled for un-gated pooled inference, PULSE_GATED_BLEND=1 (Phase 2.6) for the gated blend, and PULSE_GATED_SIZE=<full|half|kelly> (Phase 2.7) to scale the regime-leg notional. Default is composite + full sizing for back-compat with Sprint 3 dashboard cards.

• **Gate is conservative by design** — Phase 2.5 demonstrated lift only on (BACK regime × {Lasso, Huber} winners). The gated blend codifies exactly that slice; per-spread Sharpe and the regime/baseline split in the headline disclose how often the engine actually fires versus deferring to the baseline.

• **Phase 2.7 sizing has a Sharpe-vs-DD trade-off** — uniform 0.5× halves both mean PnL and max DD on the regime leg; Kelly is per-spread and adaptive but its sample is thin (97 regime fires total). Live deployment should read sized_blend_summary.kelly_per_spread_latest from the walk-forward report — the same numbers shown above — to size at inference time.

• **Next:** fold inventory + vol back in as *features* inside the pooled regression so they inform fair value without fragmenting the training set; trial 5/60-day horizons (one-line change in walkforward.py); consider per-spread gate thresholds where the per-spread lift table justifies them.